FastSpeech 2 Text-to-Speech: English (Svarah) and Telugu (IndicTTS)

Technical report by Sumit Chatterjee

Code and results: GitHub repository

Trained checkpoints: Google Drive folder

Reference implementation: ming024/FastSpeech2. Paper: Ren et al., 2021

Summary

I trained two separate FastSpeech 2 models, one for English and one for Telugu, starting from the ming024/FastSpeech2 code. Each model saw about 5 hours of training audio and about 1 hour of validation audio. Training ran for 35 epochs per language, which is roughly 4,000 optimizer updates for English and 3,850 for Telugu.

The results are poor and I want to say that up front. The English output is mostly a low hum with a faint, hazy voice behind it, and the words cannot be made out. The losses went down normally, so the problem is not visible in the loss curves alone. My current guess is a combination of three things: very short training compared with the hundreds of thousands of updates FastSpeech 2 normally gets, duration targets that were split evenly across phonemes instead of coming from a forced aligner, and a pretrained vocoder that was trained on a different speaker and language. These are hypotheses. I have not run the checks in Section 6.2 that would separate them.

1. Implementation approach

1.1 What I used from the reference code

The model follows the FastSpeech 2 paper as implemented in ming024/FastSpeech2. The parts I used are:

Component	What it does
Encoder	Symbol embedding followed by a stack of feed-forward Transformer (FFT) blocks, each with multi-head self-attention and a convolutional feed-forward layer
Variance adaptor	Duration, pitch and energy predictors. Pitch and energy are quantized into bins (kept as model buffers), embedded, and added to the hidden sequence
Length regulator	Repeats each phoneme representation for its duration in mel frames (ground truth during training, predicted at inference)
Decoder and PostNet	FFT-block decoder gives the mel spectrogram and a convolutional PostNet refines it. Both are trained with L1 loss
Loss	Mel L1 + PostNet L1 + pitch + energy + log-duration MSE, summed into one total

1.2 Changes I made

Change	File	Why
Attention mask fill value changed from -1e9 to the minimum value of the tensor dtype	core/model.py	-1e9 does not fit in float16 and training with mixed precision crashed with an overflow error. Same change for both languages
At inference, duration is clamped to a minimum of 1 and rounding is done after the control factor is applied	core/model.py	Stops a phoneme from getting 0 frames and disappearing
Predictor output biases initialised: duration 1.75, pitch and energy to the dataset means	training loop	Starts the predictors in a sensible range so early training is steadier

Change	File	Why
Mixed-precision training with GradScaler, AdamW, OneCycle learning-rate schedule	training loop	Faster training on one GPU
Pitch and energy bin buffers loaded from the checkpoint	inference.py	The bins depend on dataset statistics, so they have to match the trained model and not the defaults
Pitch from YIN with linear interpolation over unvoiced frames; energy as the L2 norm of the STFT magnitude per frame	core/audio.py	Simple frame-level targets that line up with the mel frames
Duration targets: mel frames divided evenly across the phonemes of an utterance (T_mel // N_ph)	preprocessing	I did not use forced-alignment output for durations. This is a big departure from the reference code and I think it is the most important weakness of the whole project (see Section 6)
Mel to waveform: Griffin-Lim (64 iterations) and a pretrained HiFi-GAN both implemented	core/audio.py	See 1.3

1.3 Vocoder

I implemented two ways of turning the mel into audio. The first is Griffin-Lim with 64 iterations, using the pseudo-inverse of the mel basis. The second is a pretrained HiFi-GAN (jaketae/hifigan-lj-v1, which was trained on LJSpeech). The test audio I am submitting was generated with HiFi-GAN. It sounds smoother than Griffin-Lim, which has a metallic, smeared quality, but both are limited by how blurry the predicted mels are. HiFi-GAN was trained on one English speaker, and I used the same vocoder for Telugu as well.

1.4 Telugu-specific changes

The Telugu text front end uses the Indic NLP Library tokenizer plus Morfessor to break words into smaller units, so that conjuncts (vattu/othu) and long compound words are handled. The symbol vocabulary was enlarged to 128 Unicode symbols covering Telugu

vowels, consonants, vowel signs (guninthalu) and the virama. Sampling rate, FFT size and hop length are the same as for English.

2. Data and training setup

2.1 Selection and split

	English (Svarah)	Telugu (IndicTTS Telugu)
Training utterances	3,644 (about 5.0 h)	3,520 (about 5.0 h)
Validation utterances	716 (about 1.0 h)	700 (about 1.0 h)
How the subset was picked	Streamed the dataset and took utterances in order	Took utterances in order
Train/validation overlap	None. The two sets are separate blocks of the stream	None. Separate blocks

For English I streamed the Svarah dataset (Bhargav0044/svarah1) and cut it into a training block of about 5 hours and a validation block of about 1 hour, with no utterance appearing in both. Telugu was handled the same way with the IndicTTS Telugu data. All audio was resampled to 22,050 Hz, 16-bit mono. Because the split is by position and not by speaker, the same speakers can appear in both training and validation.

2.2 Audio features

Parameter	Value
Sampling rate	22,050 Hz
STFT	FFT 1024, hop 256, window 1024, Hann
Mel	80 bins, 0 to 8000 Hz, natural log, clipped at 1e-5
Pitch	YIN, 60 to 500 Hz, unvoiced frames filled by linear interpolation
Energy	L2 norm of the STFT magnitude per frame
English statistics	mean pitch 172.7 Hz, mean energy 45.0

2.3 Training configuration

Setting	Value
Optimizer	AdamW, betas (0.9, 0.98), eps 1e-8, weight decay 1e-4
Learning rate	OneCycle, max 1.5e-3, 15% warm-up
Epochs and batch size	35 epochs, batch size 32 (114 updates per epoch for English, about 3,990 in total)
Precision	float16 autocast with GradScaler
Gradient clipping	1.0
Checkpoint choice	Lowest validation mel loss (mel and PostNet L1 averaged)
Model size	hidden size 256, 4 encoder layers, 4 decoder layers, 2 heads each, FFT inner dimension 1024, 80 mel channels

Telugu used the same settings with its own config file. The only differences are the ones in Section 1.4.

2.4 Preprocessing steps

The same steps were run for both languages, in English/preprocess.py and Telugu/preprocess.py.

1. Stream the dataset and take utterances in order: about 5 hours for training, then a separate block of about 1 hour for validation.
2. Resample every utterance to 22,050 Hz, 16-bit mono.
3. Convert the text to a symbol sequence. English becomes phoneme symbols (the duration plots show them with stress digits). Telugu goes through the Indic NLP tokenizer and Morfessor into the 128-symbol vocabulary.
4. Compute the 80-bin log-mel spectrogram with the settings in Section 2.2.
5. Compute frame-level pitch (YIN with interpolation over unvoiced frames) and energy (L2 norm of the STFT magnitude).
6. Compute dataset statistics (mean pitch and energy), which set the predictor biases and the pitch and energy bins.
7. Make the duration targets by dividing the mel frames of each utterance evenly across its phonemes.

3. Loss curves

3.1 English

Metric	Epoch 1 (train / val)	Epoch 35 (train / val)	Best val epoch
Mel L1	2.86067 / 1.63141	0.58387 / 0.87785	16 (0.85202)
Duration	0.27341 / 0.10163	0.01866 / 0.05693	
Pitch	0.58443 / 0.5841	0.39595 / 0.61458	
Energy	2.62415 / 2.4773	0.92353 / 2.95867	
Total	9.20333 / 6.42585	2.50588 / 5.38588	

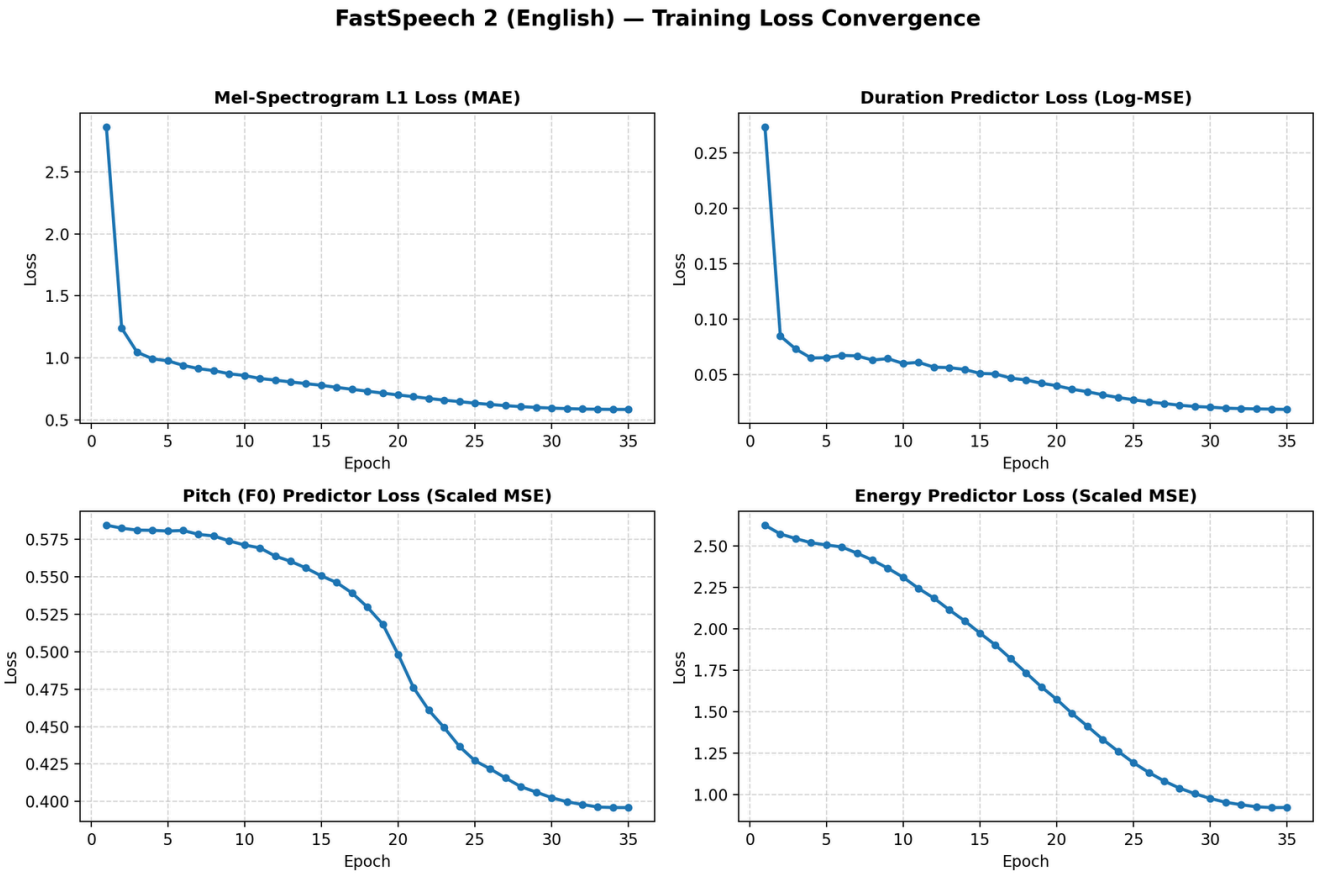


Figure 1. English: training loss curves (mel, duration, pitch, energy).

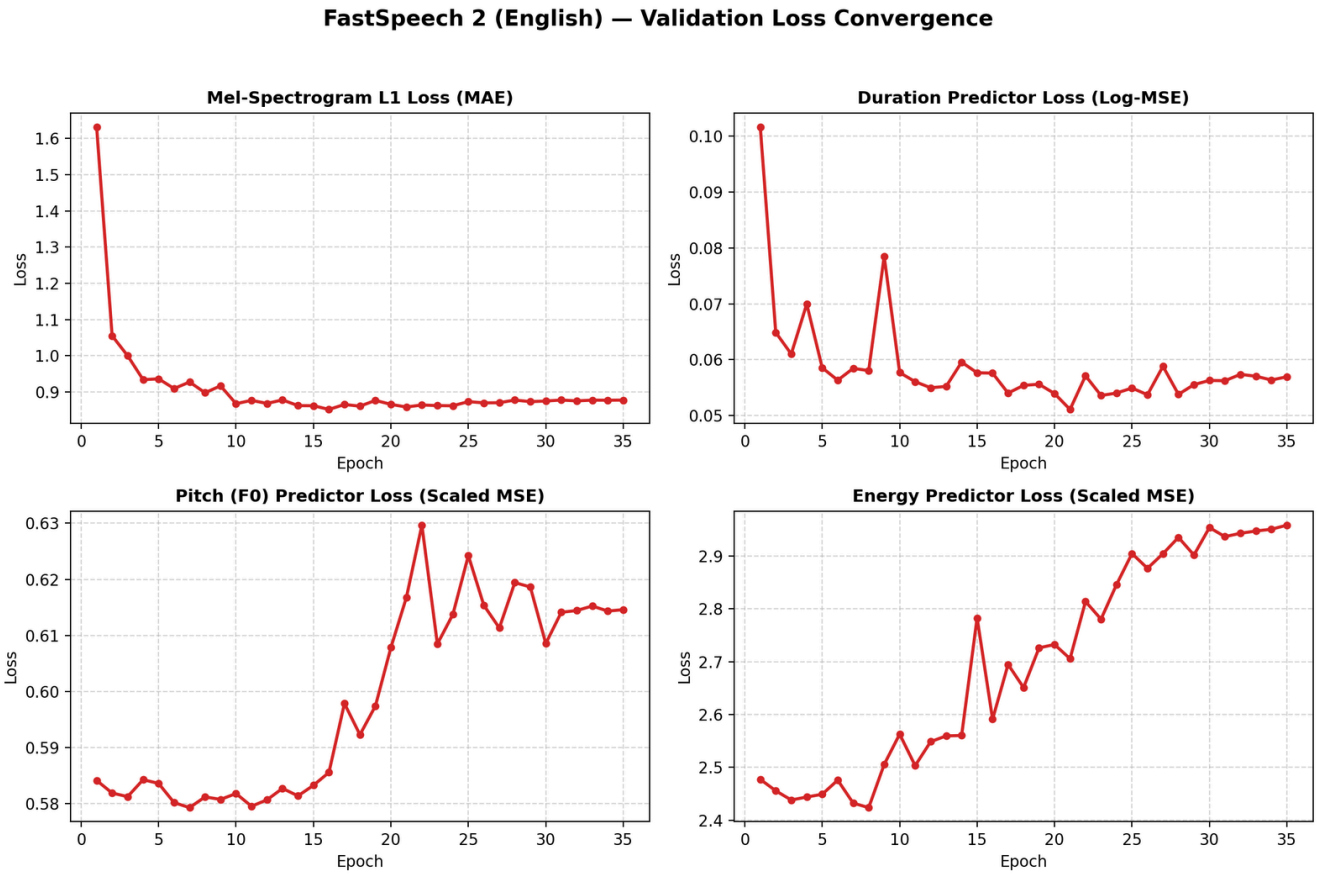


Figure 2. English: validation loss curves.

Training mel loss fell steadily from 2.861 to 0.584, a drop of 79.6%. Validation mel loss dropped quickly to about 0.9 within the first five or six epochs, reached its lowest value of 0.852 at epoch 16, and then crept back up to 0.878 by epoch 35. That small rise is the first sign of overfitting.

The pitch and energy predictors show it much more clearly. Their training losses kept falling (pitch down 32.3%, energy from 2.62 to 0.92), but their validation losses went the other way after about epoch 10. Validation pitch loss rose from about 0.58 to 0.61 and validation energy loss rose from about 2.5 to 2.96, which ends higher than where it started. Validation duration loss stayed noisy around 0.05 to 0.06 with one spike near epoch 9. Nothing diverged and there were no gradient problems, but the gap between training total (2.51) and validation total (5.39) shows the model is fitting the training set much better than the validation set.

3.2 Telugu

Metric	Epoch 1 (train / val)	Epoch 35 (train / val)	Best val epoch
Mel L1	3.39214 / 1.98363	0.45703 / 0.6904	20 (0.68395)
Duration	0.20993 / 0.03602	0.01617 / 0.0157	
Pitch	0.4814 / 0.42296	0.27029 / 0.47915	
Energy	2.28976 / 1.84238	0.5465 / 2.25352	
Total	9.76536 / 6.26864	1.747 / 4.12917	

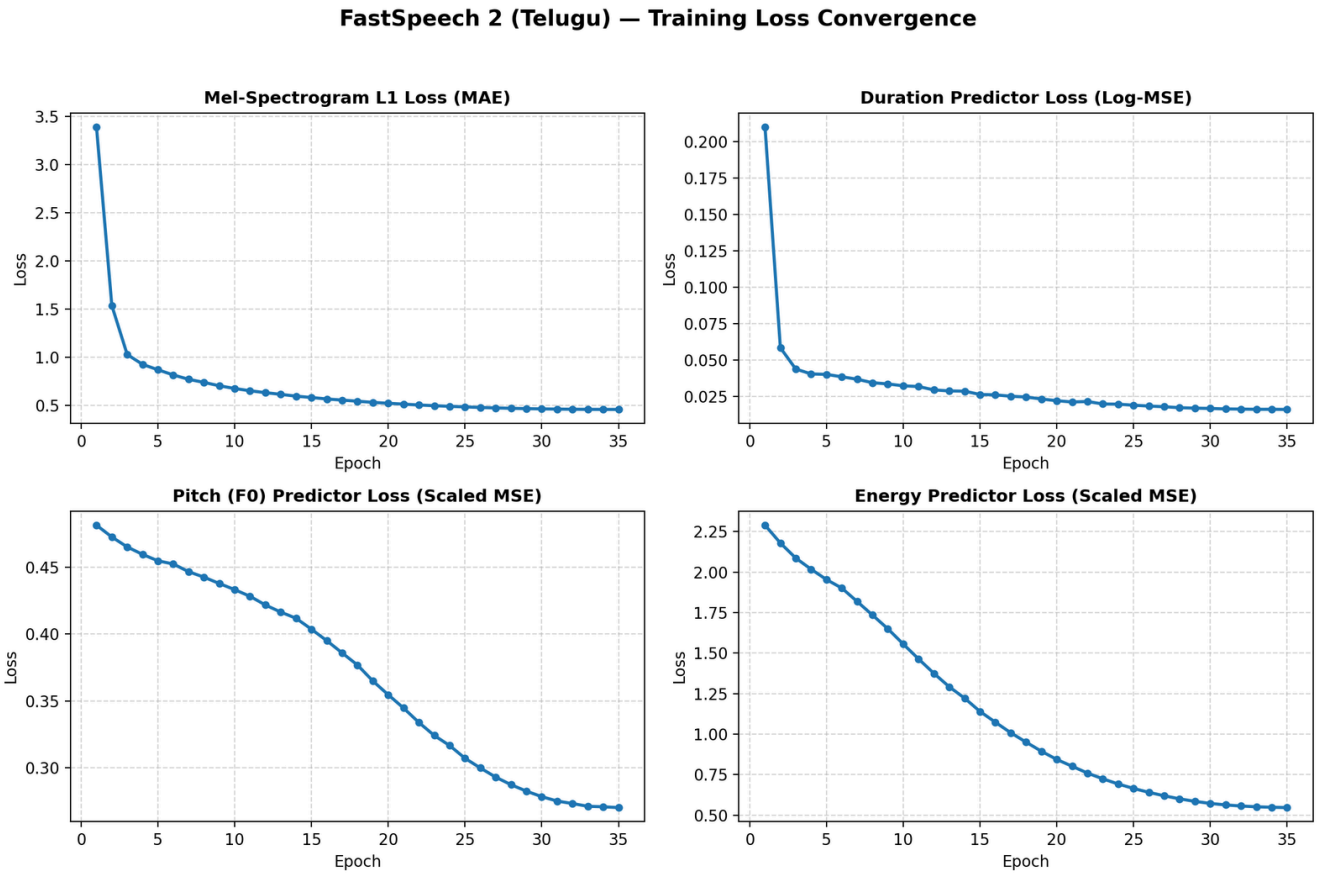


Figure 3. Telugu: training loss curves (mel, duration, pitch, energy).

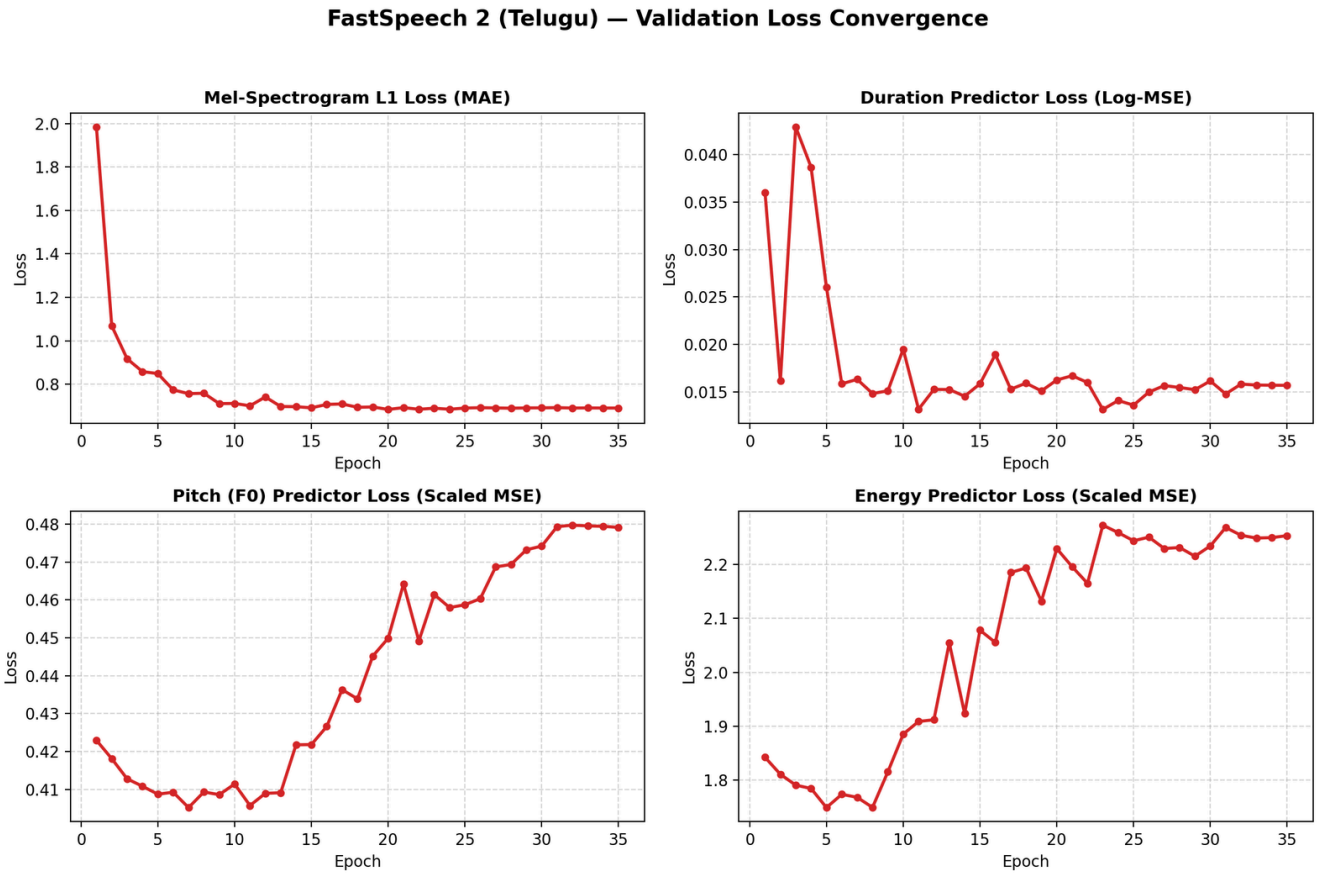


Figure 4. Telugu: validation loss curves.

Telugu training mel loss went from 3.392 to 0.457 (down 86.5%). Validation mel loss reached 0.684 at epoch 20 and then stayed almost flat to the end (0.690 at epoch 35). Validation duration loss was spiky in the first few epochs and then settled around 0.015 to 0.016.

As with English, the pitch and energy predictors overfit. Validation pitch loss was lowest at about 0.41 around epoch 6 and climbed to 0.479 by the end, while training pitch loss dropped 43.9%. Validation energy loss went from about 1.75 at its lowest to 2.25. So in both languages the mel loss flattens out while the variance predictors keep fitting the training data and get worse on validation data.

One thing worth noting for both languages: the validation mel loss has been flat for the last 15 or so epochs. Simply training longer on the same 5 hours would probably not help much on its own. Better duration targets and more data look more promising than more epochs.

4. The generated speech

Audio is in Results/<Language>/test_generated_audios/. The test sentences are used exactly as given in the assignment, including the 'rhyth m' with the space in E3.

English overall. The output is mostly a low hum with a hazy, barely audible voice behind it, and the words cannot be recognised. This is true for all five English files, so I am not claiming any sentence is intelligible.

4.1 English sentences

ID	Sentence	Length	Comment
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.	8.94 s	The longest word in the set. The audio is a long stretch of low-frequency hum, so the individual consonants in the long word cannot be told apart
E2a	The orchestra's conduct during the ceremony was praised by the audience.	5.57 s	'conduct' used as a noun. Pronunciation cannot be judged from the audio
E2b	The researchers will conduct an experiment to evaluate the proposed method.	6.32 s	'conduct' used as a verb. I did not add anything to the front end to choose between the noun and verb reading, so I would not expect the stress to differ between E2a and E2b
E3	The rhyth m of the orchestra changed unexpectedly.	4.11 s	Passed with the space as given, so 'rhyth' and 'm' are treated as two separate words
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.	8.36 s	No text normalisation, so the digits and the date are not turned into words. 'andaroo' is not a dictionary word and goes through the character fallback. Both come out as hum

4.2 Telugu sentences

The Telugu sentences are referred to by ID (T1, T2a, T2b, T3, T4) as printed in the assignment PDF.

ID	What it tests	Length	Comment
T1	Long technical compound (pharmacological research)	4.95 s	Longest single word in the Telugu set
T2a	Homograph, 'grip' on a rope	2.57 s	Same spelling as T2b but a different meaning
T2b	Homograph, silk saree	3.29 s	Longer than T2a because the sentence has more symbols
T3	Short sentence with soft conjunct consonants (the tiger walked silently in the forest)	2.41 s	Shortest of the Telugu test files
T4	Date, time and numerals	8.72 s	Same kind of test as E4. No number normalisation

[Sumit: add a line or two here about what the Telugu audio actually sounds like, in the same plain way as the English paragraph above. I have not written any listening notes for Telugu because the original draft did not have an overall one.]

5. Variance adaptor plots

Plots for each sentence are in Results/<Language>/variance_adaptor_plots/ under duration, pitch and energy.

5.1 What the plots show in general

Duration. In both languages the predicted durations fall into just a few repeated values and do not vary smoothly. English values sit mostly around 5, 7 and 13 to 14 frames, with a few up to 17 or 18. Telugu values are lower and tighter, between about 3 and 9 frames. This is what you would expect if the model was trained on targets that were split evenly across phonemes, because then there is very little real timing information to learn. It also fits the very low duration loss (about 0.017), which is lower than I would expect from real alignments.

Pitch. English contours mostly stay between about 160 and 195 Hz and slope downward across the sentence, which is close to the training mean of 172.7 Hz. Telugu contours are higher and swing much more, from about 120 up to nearly 290 Hz.

Energy. Almost every plot has a sharp burst in the first fraction of a second (around 75 to 85 for English, 100 to 127 for most Telugu sentences) before settling down. I think this comes from how the targets were computed at utterance starts, but I have not checked.

5.2 English

- **E1** (8.94 s): pitch stays mostly between 165 and 190 Hz and drops to about 145 Hz at the very end. Energy bursts near the start and then moves between about 30 and 70.
- **E2a** (5.57 s, 57 phonemes): pitch between about 160 and 194 Hz with a high peak near 4.4 s. Energy has three clear peaks, at about 0.3 s, 3.1 s and 4.5 s.
- **E2b** (6.32 s): pitch between about 149 and 196 Hz with peaks near 2.3 s and 3.4 s. Energy peaks near 0.6 s, 3.7 s and 6.0 s.
- **E3** (4.11 s): pitch slopes down from about 180 Hz to about 145 Hz near 3.7 s and then rises a little. Energy starts high and decays over the sentence. I could not see a distinct break in any of the three plots at the 'rhyth m' split.
- **E4** (8.36 s, 98 phonemes): pitch between about 155 and 196 Hz with no clear step at the date and time. Energy falls to a low stretch around 6.5 to 7 s.

Figures 5 to 9 show the predicted duration (top), pitch (bottom left) and energy (bottom right) for each English test sentence.

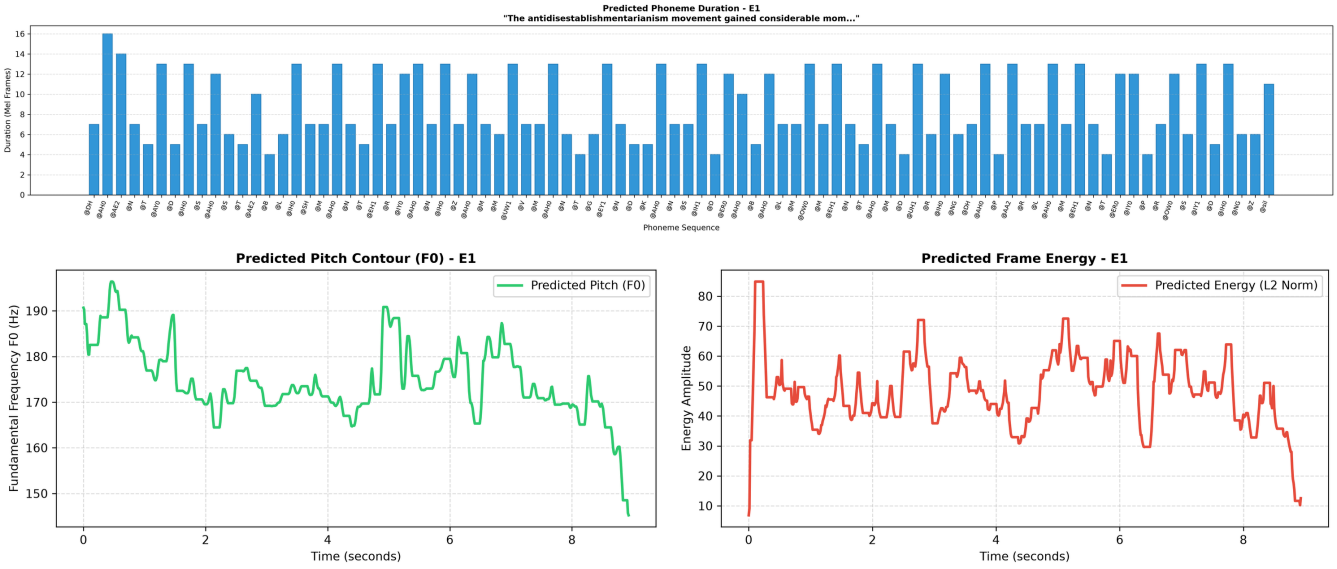


Figure 5. English E1: predicted duration, pitch (F0) and energy.

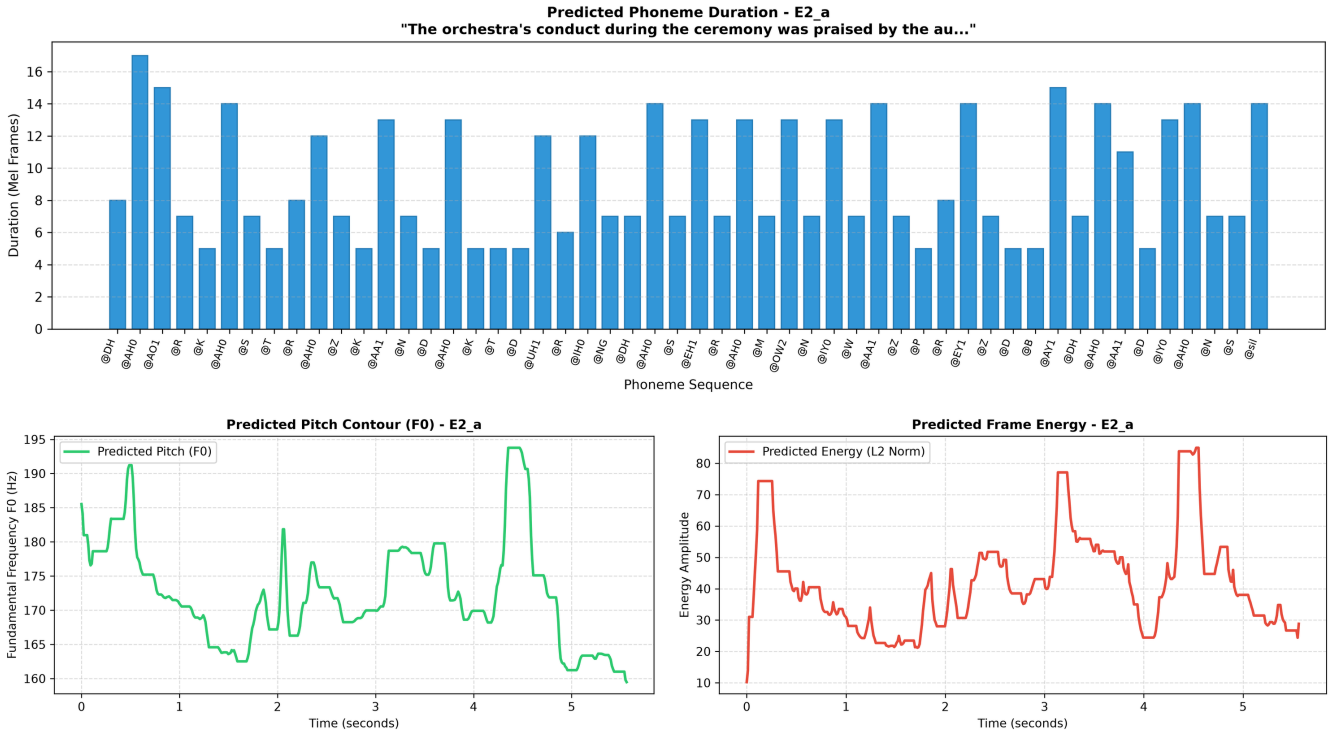


Figure 6. English E2a: predicted duration, pitch (F0) and energy.

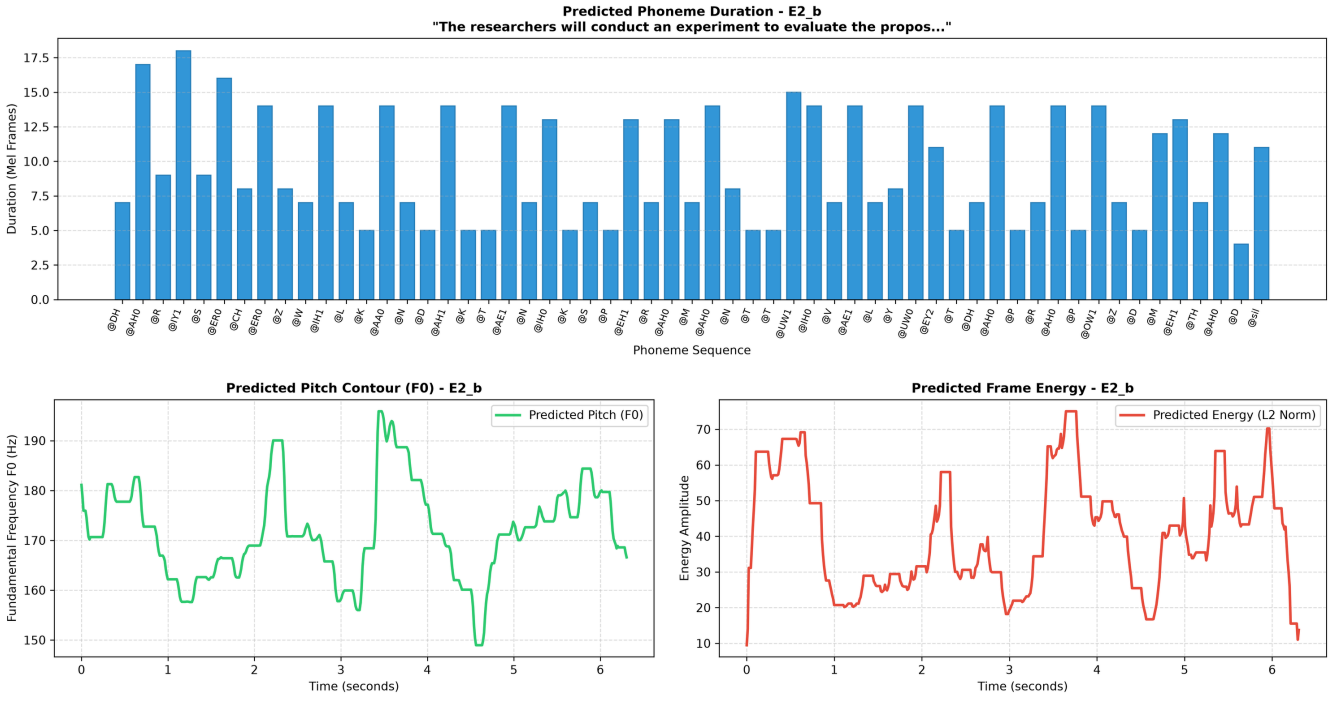


Figure 7. English E2b: predicted duration, pitch (F0) and energy.

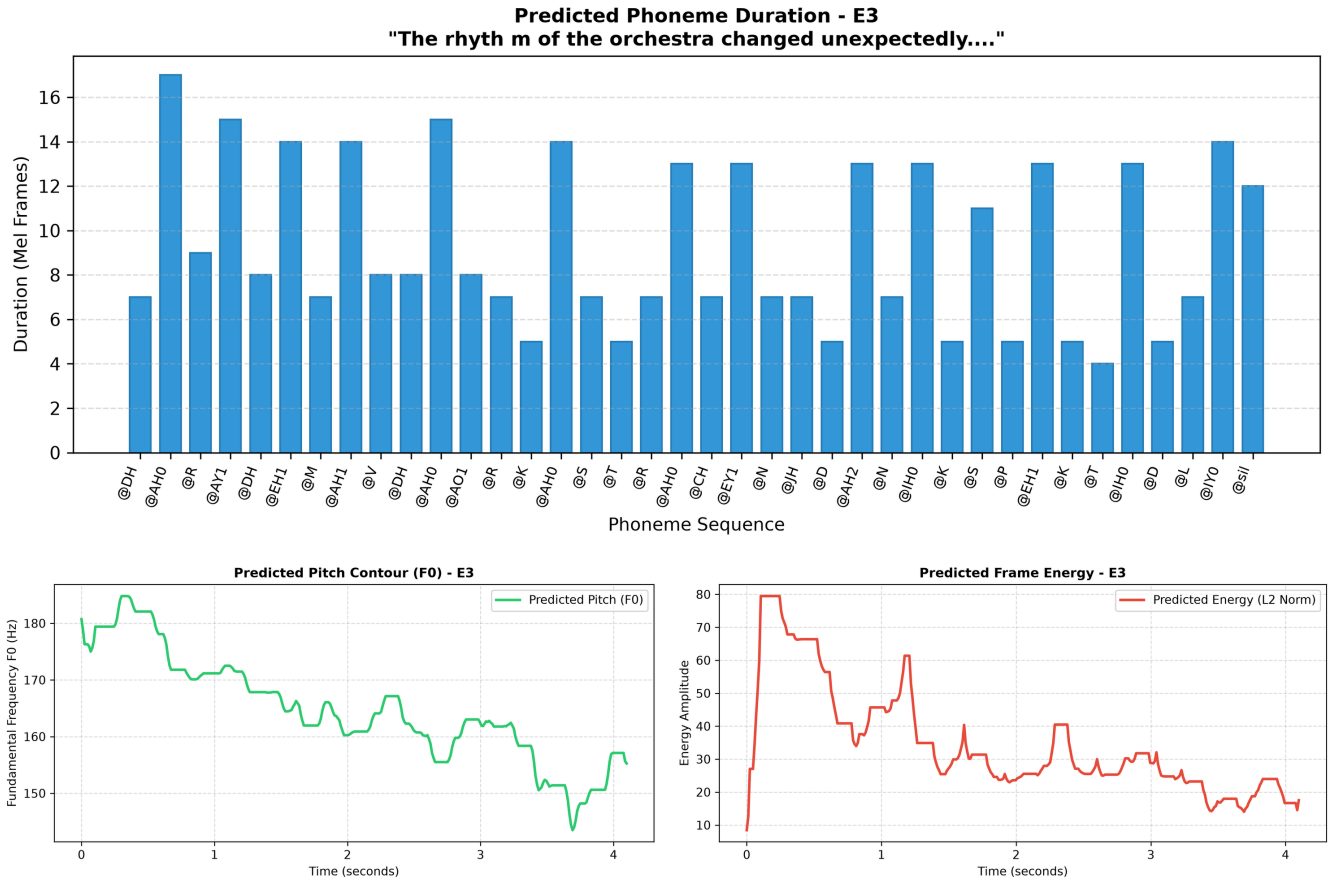


Figure 8. English E3: predicted duration, pitch (F0) and energy.

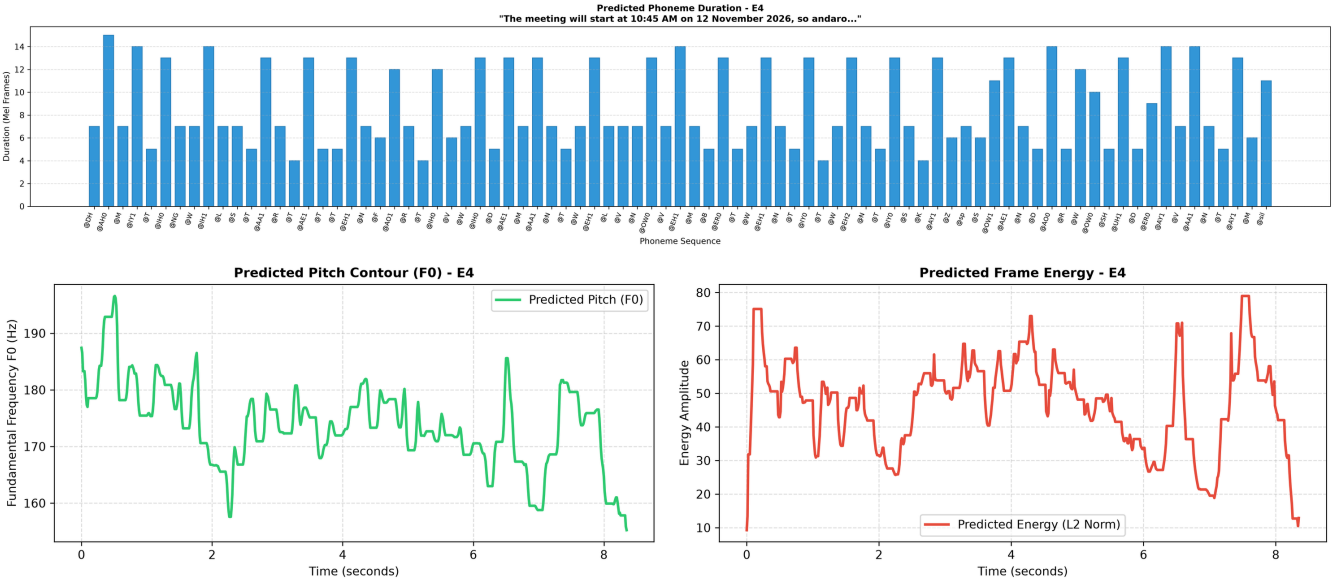


Figure 9. English E4: predicted duration, pitch (F0) and energy.

Reading the English plots. Pitch drifts downward over almost every sentence and finishes low, and it stays inside a narrow band of roughly 150 to 195 Hz, close to the training mean of 172.7 Hz. This is what a predictor tends to do when it has learned the

average contour and not word-level accents, and it is consistent with the pitch validation loss rising in Figure 2. The sharp energy rise in the first fraction of a second appears in every English sentence. E2a and E2b look alike in the same way: the two uses of 'conduct' get no visibly different treatment. In E4 the digits and the date produce no step in pitch or energy.

5.3 Telugu

- **T1** (4.95 s): pitch from about 125 to a peak near 290 Hz early on, mostly 150 to 220 Hz after that. Energy has many peaks and dips with a high burst at the start.
- **T2a** (2.57 s): pitch between about 137 and 260 Hz, falling toward the end of the sentence. Energy has several separate peaks.
- **T2b** (3.29 s): pitch between about 156 and 257 Hz with an early peak. Energy starts high and tapers off.
- **T3** (2.41 s): pitch starts near 285 Hz, falls to about 123 Hz around 1 s, rises again and ends low. Energy starts at about 127 and falls through the sentence.
- **T4** (8.72 s): pitch is jagged, between about 120 and 255 Hz, with many short peaks. Energy goes close to zero around 6.8 to 7.2 s, which may be the comma, and has a final peak near 8.2 s.

Figures 10 to 14 show the same three plots for the Telugu test sentences: duration (top), pitch (bottom left) and energy (bottom right).

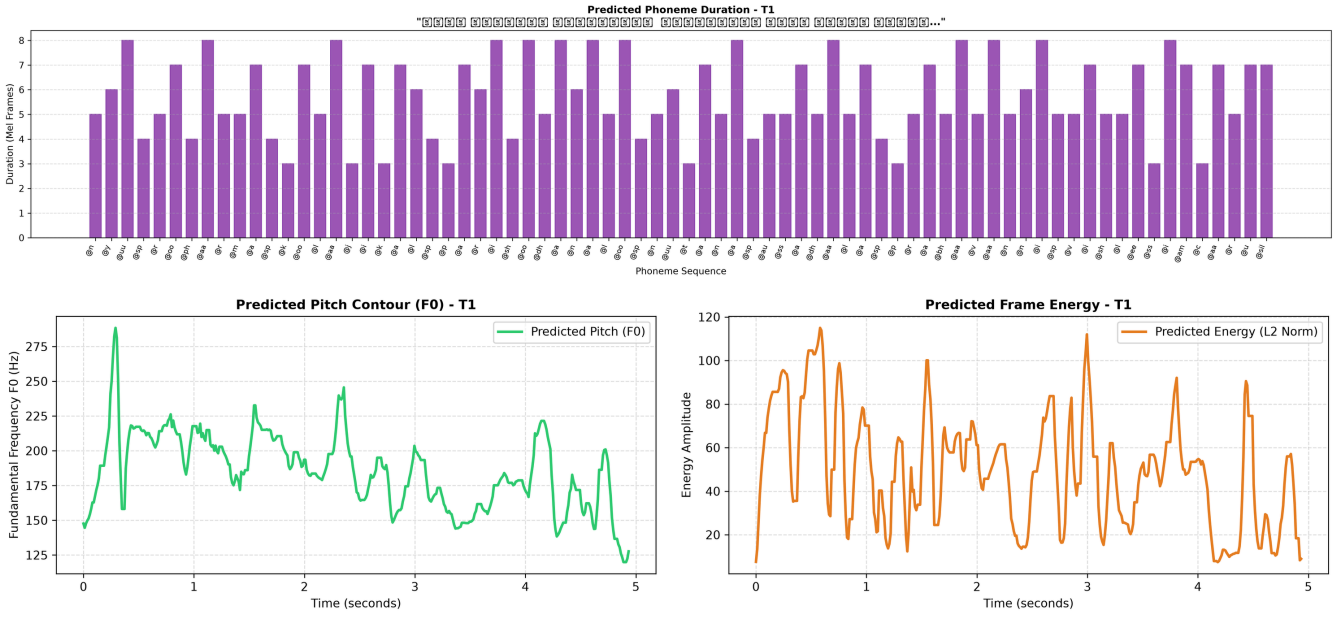


Figure 10. Telugu T1: predicted duration, pitch (F0) and energy.

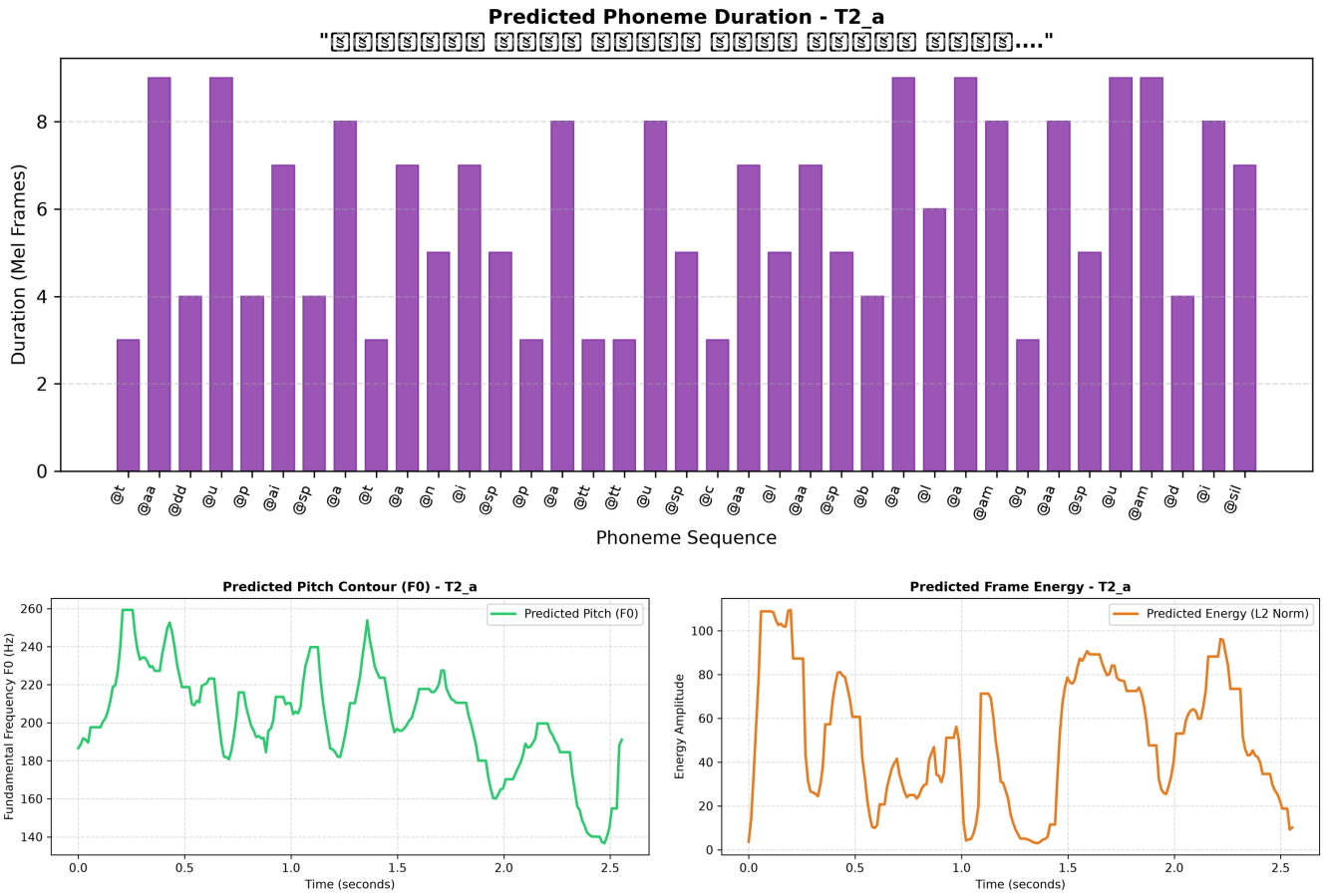


Figure 11. Telugu T2a: predicted duration, pitch (F0) and energy.

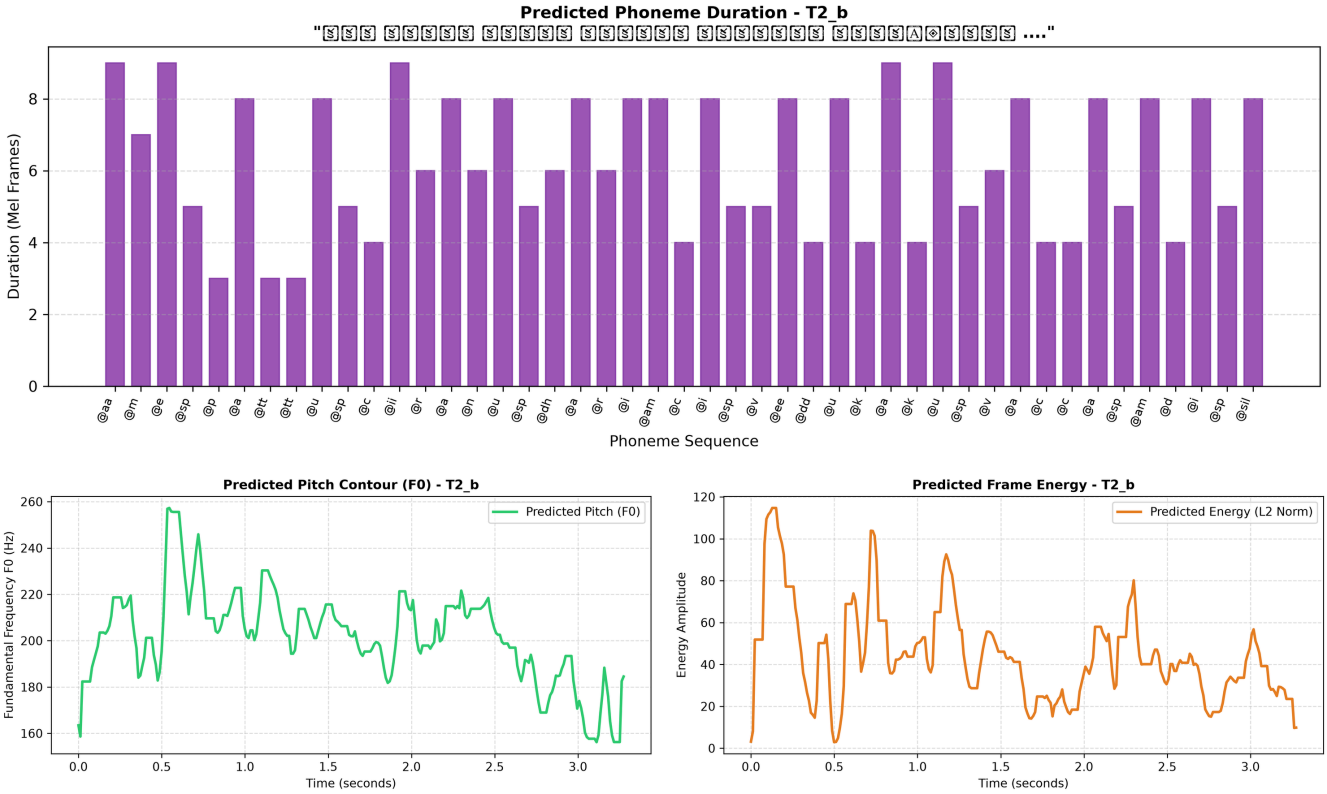


Figure 12. Telugu T2b: predicted duration, pitch (F0) and energy.

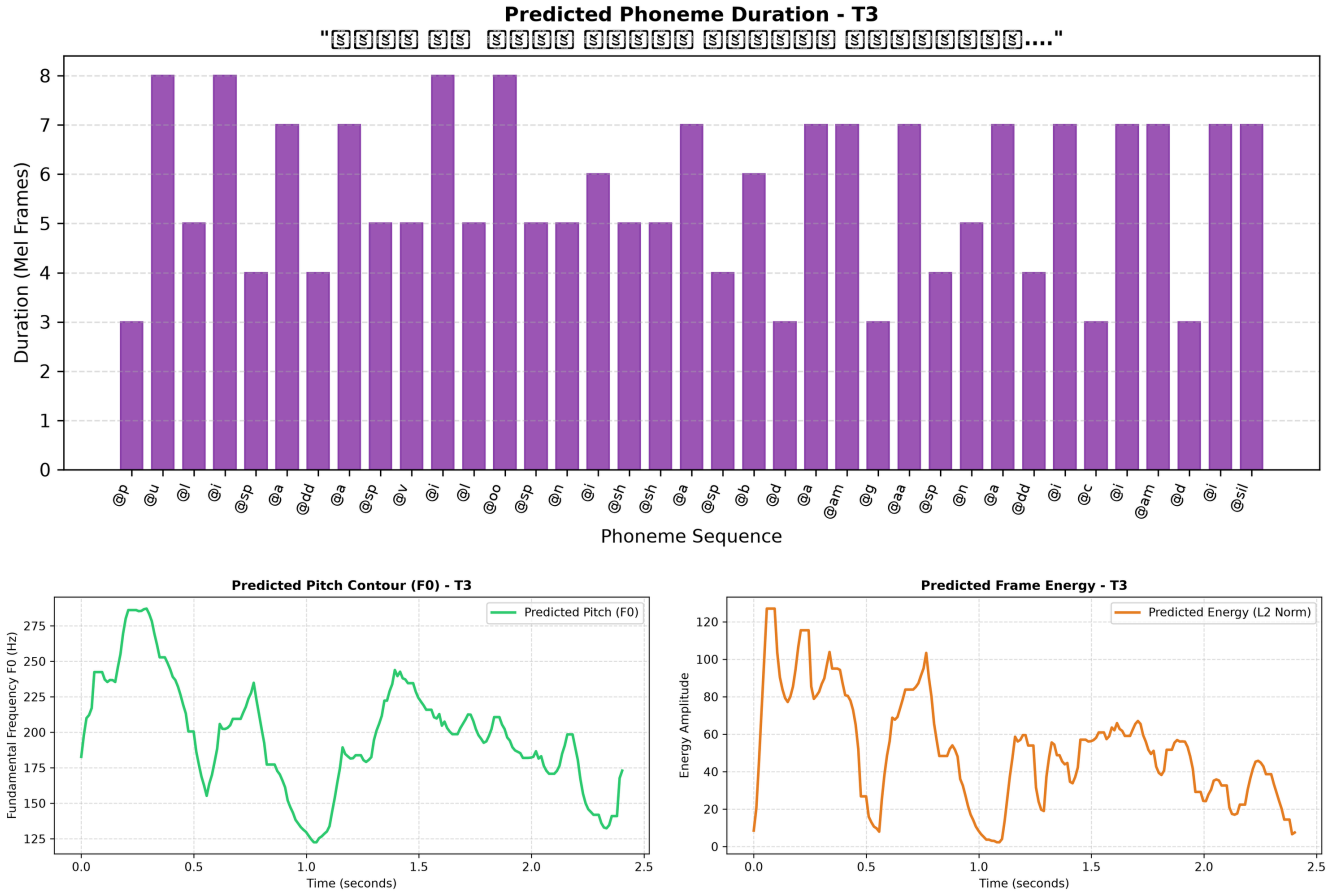


Figure 13. Telugu T3: predicted duration, pitch (F0) and energy.

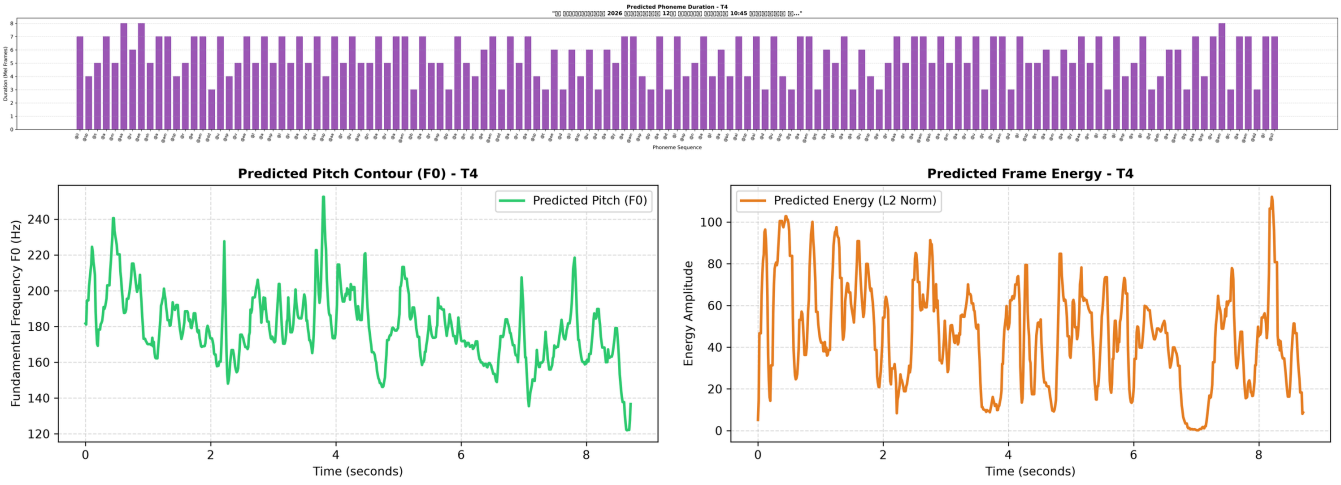


Figure 14. Telugu T4: predicted duration, pitch (F0) and energy.

Reading the Telugu plots. The Telugu pitch swings are much larger than the English ones, with peaks near 290 Hz in T1 and T3 and dips down to about 120 Hz. I did not compute Telugu pitch statistics for this report, so I can't say how these compare with the training

mean. As in English, energy starts with a sharp burst and then settles. In T4 energy falls close to zero around 7 s, which may be the comma in the sentence, while the date and time themselves show no clear change.

6. Comparing the two models, limitations and next steps

6.1 English and Telugu side by side

	English	Telugu
Training mel loss, epoch 35	0.584	0.457
Best validation mel loss	0.852 (epoch 16)	0.684 (epoch 20)
Training duration loss, epoch 35	0.0187	0.0162
Validation pitch loss, epoch 35	0.615	0.479
Validation energy loss, epoch 35	2.959	2.254

Telugu reached lower mel and variance losses than English. I can think of two reasons. Telugu spelling maps to sounds much more regularly than English spelling does, and the two datasets are different, so the numbers are not a fair like-for-like comparison. Despite the lower numbers, the failure is the same in both languages: smoothed-out mels and a hum in the audio. That points at something shared between the two pipelines, such as the duration targets, the amount of training, or the vocoder, and away from anything specific to English or Telugu.

On the test sentences. The two sets test the same four things (a long word, a homograph pair, a short sentence, and a date with a time), and the plots treat them alike in both languages. The long-word sentences E1 and T1 give long stretches with no visible boundary at the word. In both homograph pairs (E2a/E2b and T2a/T2b) the length difference follows from how many symbols each sentence has, and nothing in the plots suggests the model separated the two meanings. The date and time sentences E4 and T4 both run about 8.4 to 8.7 s and show no clear change at the numbers, which is what I would expect without text normalisation. The short sentences E3 and T3 are the cleanest cases, but E3 shows no visible break at the 'rhyth m' split, and T3 has the largest pitch swing of the Telugu set (from about 285 Hz down to about 123 Hz in 2.4 s).

6.2 Limitations

1. **Duration targets.** Frames were divided evenly across phonemes (T_mel // N_ph) and not taken from a forced aligner. Phoneme boundaries therefore do not line up with the real audio, and the decoder learns to predict a blurry average spectrum. I think this is the main reason the output is a hum.
2. **Training length.** About 4,000 updates per model is far below what FastSpeech 2 normally needs (the usual figure is over 100k).
3. **Vocoder.** The pretrained HiFi-GAN was trained on one English speaker (LJSpeech). It was not trained on Telugu or on Indian-accented English.
4. **Text front end.** There is no number, date or time normalisation, so E4 and T4 cannot be pronounced properly. Unusual words like 'andaroo' go through a character fallback.
5. **Overfitting of pitch and energy predictors.** Validation loss for both rises while training loss falls, in both languages.
6. **Mixed speakers.** The train and validation sets were split by position and not by speaker, and I did not normalise pitch and energy per speaker.

6.3 Checks I have not run yet

1. Vocode a ground-truth mel with the same vocoder. If that also sounds bad, the vocoder or the mel settings are the problem and not the model.
2. Plot a predicted mel next to the ground-truth mel for one validation sentence. A smeared mel with no clear harmonics points to an under-trained model or bad alignment.
3. Compare the value ranges of real and predicted mels, to rule out a normalisation mismatch.

6.4 What I would try next

- Get proper forced alignments for the duration targets and re-run preprocessing.
- Train for many more updates, resuming from the best checkpoint with a lower peak learning rate.
- Normalise pitch and energy per speaker, or use a single-speaker subset.
- Add text normalisation for numbers, dates and times, and a better grapheme-to-phoneme step for unusual words and Telugu conjuncts.
- Try a vocoder that matches the data better, and check that its mel settings match mine exactly.

7. Reproducibility

7.1 Software

Python 3, PyTorch (CUDA build for training), librosa, soundfile, scipy, numpy, PyYAML, matplotlib, tqdm and speechbrain (for the pretrained HiFi-GAN). Exact versions are in requirements.txt.

7.2 Commands

Preprocessing:

```
python English/preprocess.py
python Telugu/preprocess.py
```

Training (35 epochs each):

```
python English/train.py --config English/config.yaml --epochs 35
python Telugu/train.py --config Telugu/config.yaml --epochs 35
```

Inference:

```
python English/inference.py --config English/config.yaml --checkpoint
English/checkpoints/best_model.pth
python Telugu/inference.py --config Telugu/config.yaml --checkpoint
Telugu/checkpoints/best_model.pth
```

Config files are English/config.yaml and Telugu/config.yaml. The full training history for each language is saved in Results/<Language>/training_history.json.

Code: github.com/LightningHat/FastSpeech-2-Based-Text-to-Speech-Synthesis

Checkpoints: Google Drive folder